Team Control Number

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Problem Chosen

B

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Summary Sheet

This paper develops a comprehensive mathematical model to evaluate environmental sustainability for selecting optimal host cities for major sporting events, using the NFL Super Bowl as a primary case study. We first identified a set of key, quantifiable sustainability indicators spanning five critical domains: energy consumption, emissions, water usage, waste management, and ecological sensitivity. These indicators were subsequently normalized to facilitate comparative analysis across diverse urban environments. We constructed a robust hybrid multi-criteria decision-making (MCDM) framework that integrates the Analytic Hierarchy Process (AHP) with the Entropy Weight Method. This hybrid approach effectively combines subjective expert judgment with objective data dispersion to determine more balanced and reliable criterion weights. The derived weights were then employed within the TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) method to generate a final ranking of candidate cities. When applied to a selection of six U.S. cities, the model identified Las Vegas as the top-ranked candidate, primarily attributable to its superior energy infrastructure and relatively low ecological sensitivity. Among cities without prior Super Bowl hosting experience, Jacksonville emerged as the highest-ranked option. To assess its broader applicability, the model was successfully extended to a FIFA World Cup hosting scenario. In this global context, after incorporating accessibility as an additional crucial criterion, Beijing was identified as the top-performing city. A sensitivity analysis was conducted, which confirmed the model's robustness and its relative insensitivity to minor parameter variations. In conclusion, this research provides event organizers and policymakers with a scalable, transparent, and adaptable decision-support tool. The framework enables a more scientifically-grounded and sustainable approach to event planning, effectively balancing complex operational requirements with critical environmental considerations across different event types and scales.

Keywords: Environmental Sustainability; Super Bowl; Host City Recommendation; Multi-Criteria Decision Making (MCDM); Analytic Hierarchy Process (AHP); Entropy Weight Method (EWM); TOPSIS; Sensitivity Analysis.

Contents

1 Introduction **1**

1.1 Background . . . 1

1.2 Questions Restatement . . . 1

2 Assumptions and Justification **2**

3 Methodology **3**

3.1 Problem understanding (Task 1) . . . 3

3.1.1 Analysis of areas that impact the sustainability of Super Bowl LIV . . . 3

3.1.2 Changes of impacts in other locations . . . 3

3.2 Variables . . . 5

3.2.1 Impact on the event: Energy structure . . . 5

3.2.2 Impact on the event: Carbon emission . . . 6

3.2.3 Urban Vulnerability: Waste treatment Capacity . . . 6

3.2.4 Urban Vulnerability: Environmental sensitivity . . . 6

3.2.5 Urban Vulnerability: Resource structure . . . 6

3.2.6 Normalization . . . 7

3.3 Modeling (Task 2) . . . 8

3.3.1 Overall structure . . . 8

3.3.2 AHP . . . 8

3.3.3 EWM . . . 9

3.3.4 TOPSIS . . . 9

3.4 Model Application (Task 3) . . . 11

3.4.1 Application 1: City Recommendation . . . 11

3.4.2 Application 2: Considering Three More Cities . . . 12

3.5 Model Expansion: World Cup (Task 4) . . . 13

3.5.1 Model Revision . . . 13

3.5.2 Revised Model Application . . . 14

3.5.3 Strategies Analysis . . . 15

3.5.4 Main differences in sustainability impacts . . . 16

3.6 Sensitivity Analysis . . . 17

3.7 Strategies and their improvements on cities . . . 18

3.8 Advantages and Limitations . . . 19

3.8.1 Advantages . . . 19

3.8.2 Limitations . . . 19

4 Letter to NFL leadership(Task 5) **21**

1 Introduction

1.1 Background

Today, because of a fast-paced life, people are experiencing an increasing demand for entertainment, which includes sports events and competitions that have become popular recently. For example, in 2022, the Qatar World Cup attracted 3.4 million field audiences, while having the attention of 5 billion people worldwide. The large scale of competition and audience generates challenges for the organizers, which not only include economic pressure and fan capacity, but also are related to environmental sustainability. Research shows that several factors, such as energy consumption and water usage, determine the environmental sustainability and footprint. To ensure sustainability, event organizations need to establish a scientific evaluation model to prioritize environmental factors in the "pre-planning" stage. Under this condition, the Super Bowl, which is one of the world's most prominent sports mega-events that is held every year, provides an excellent example for relative research.

Every year, the Super Bowl attracts millions of people to its host city. However, while gaining tremendous social and economic value, the event is also bringing serious environmental pressure, mainly due to aspects such as traffic, accommodation, energy consumption, and other related factors. In 2025, the recent event was held in New Orleans, which consisted of the huge consumption of approximately 28 megawatt-hours of electricity, enough to support tens of thousands of U.S families during the duration of the game. This extreme energy demand, combined with the massive traffic and accommodation of hundreds of thousands of attendees, underscores that the environmental footprints of large-scale events are substantial currently.

Therefore, we aim to develop a mathematical model by considering environmental factors to help the organization establish a solution to determine which cities should host the next Super Bowl. In addition, this model should be capable when expanding to other mega-events.

1.2 Questions Restatement

Problem 1: Investigate the environmental impact What are the areas that impact the sustainability of the Super Bowl? How does using these help your team gain an understanding of the true environmental footprint of this event? How might these impacts differ in other locations? How may the environmental impact change based on the location of the event?

Problem 2: Develop an evaluation model Create a model to determine where to hold the next Super Bowl based completely on environmental concerns.

Problem 3: Model validation Determine which city should host the Super Bowl in 2029. Apply the model to a set of potential host cities within the United States and determine which would be the most environmentally sustainable choice. In addition, Apply the model to determine which potential city that has NFL teams and has never hosted would be the most environmentally sustainable choice?

Problem 4: Model extension and analysis reflection Extend the model to account for other types of sport mega-events and apply the model for one of the large-scale sporting events, and recommend the appropriate host location, discuss the optimal strategies, and compare their differences.

Problem 5: Recommendation of the host city Write a one-to-two-page letter to the NFL recommending the chosen host city for the Super Bowl in 2029.

2 Assumptions and Justification

Assumptions and Justification To simplify the problem and quantitatively evaluate the most sustainable city for the Super Bowl, we make the following assumptions:

Assumption 1: Obtaining data through population-weighting. The data of the city is assumed to be obtained by population-weighted total data of the state.

Justification: Since the information of a specific state in the US, compared to a city's information, can be gained more easily, while having a relatively high precision. Usually, the resources, pollution, and other factors we consider can be seen as evenly distributed among people, so using the population-weighted approach can be a close estimation to reality.

Assumption 2: Equal environmental impact as the last event. In the model, all the environmental impacts of the Super Bowl are considered the same as Super Bowl LIX.

Justification: Super Bowl LIX is the most recent event. Nowadays, as the energy structure is changing and the technology level is continuously increasing, the impact of mega-events is expected to change significantly. Therefore, the most recent year of the last Super Bowl can serve as a great example for the next event.

Assumption 3: Waste recovery rate represented by the median We take the median of the waste recovery rate within the interval to represent the waste recovery rate as a whole.

Justification: The waste recovery rate for a city is given by a range. However, it is easier to analyze when the rate is a specific number. Therefore, using the median of the interval is a reasonable way to estimate the rate.

Assumption 4: Restricted modes of public transportation When discussing public transportation, we typically consider only subways and public buses.

Justification: Subways and buses are the most common public transportation in big cities. Usually, various passengers take buses or subways in their everyday lives, proving that it is reasonable for us to consider only these two aspects of public transportation.

Assumption 5: Similarities in large-scale American football matches' data The Super Bowl event has a similar characteristic and impact on other American football events that demonstrate the same significant scale.

Justification: Usually, a large sports event would attract the same group of fans from all over the country as well as worldwide, while the local audience number sticks to a certain level, optimistically. Thus, the scale, consumption, and other aspects are usually similar to each other.

Assumption 6: Representing the level of air pollution by the concentration of PM2.5 We assume that the concentration of PM2.5 in the atmosphere can represent the level of air pollution, which is positively related to the intensity of the pollution source, citizens' health risks, etc.

Justification: PM2.5 is not only a source of air pollution that is harmful to the human body, but is also usually derived from the incomplete combustion of fossil fuels. It can cause situations like haze weather. Therefore, the more concentrated of PM2.5, the more pollution a city generates.

3 Methodology

3.1 Problem understanding (Task 1)

3.1.1 Analysis of areas that impact the sustainability of Super Bowl LIV

Many factors can determine the sustainability of Super Bowl LIV. Therefore, to ensure the analysis of the aspects that impact the sustainability of Super Bowl LIV is accurate, we make the table as follows, which contains four aspects that mainly determine the sustainability of Super Bowl LIV.

Table 3.1: Sustainability Metrics for Super Bowl LIX in New Orleans

Category	Value
Waste Utilization	
Material Recovery	Tens of thousands of pounds
Food Recovery	140,000 pounds
Energy Consumption	
Venue Operations (AC, Audio, Lighting)	10,500 kWh/game
Broadcasting Equipment (Screens, Satellite)	6,250 kWh/game
Greenhouse Gas Emissions	
Venue Facilities	60t CO_2
Attendee Travel	34,000t CO_2
Water Usage	
Turf Maintenance	397,500 L/game
Attendee Usage (Big Flush)	840,000 L/game

The table above shows the environmental impact of this large event. First, greenhouse gas emissions, primarily represented by the amount of CO_2, are mainly caused by attendee travel and are less related to the facilities, which form a specific level of environmental pressure. If the city can not sustain a lot of greenhouse gas emissions, holding such an event would definitely have significant and negative impact on the city. Additionally, water usage and energy consumption are very high. Specifically, the high water usage is mostly due to human activities, suggesting that the massive event could strain local water resources if the local water supply is tight. The actual pressure of the water resource may be lower than we imagine in New Orleans due to local wet conditions. Nevertheless, waste management is effective, helping to slow down environmental damage, but the overall footprint is trending negatively and is mainly driven by personal movement and gatherings.

3.1.2 Changes of impacts in other locations

Due to the precondition, our team is considering that in other cities, all kinds of information about hosting the Super Bowl is the same as that in New Orleans, such as greenhouse gas emissions and waste utilization. However, in reality, the numbers, such as waste utilization and energy consumption, are different in different areas. Besides, if the numbers are the same, different cities would face their own situations, meaning that they are facing various degrees

of environmental pressure in different aspects. Therefore, we will discuss the changes in the impacts of the event in other locations, and the following table provides information on other cities.

Table 3.2: Sustainability Comparison of Super Bowl Host Cities

City	Climate	Geography & Biodiversity	Public Transportation	Energy Mix	Water Usage	Waste Management
Santa Clara, CA	Mild, Mediterranean; moderate energy demand	Low ecological sensitivity; urban environment	Well-connected (BART, Caltrain)	High renewable share (solar, wind)	Moderate water availability	Advanced recycling infrastructure
Atlanta, GA	Warm, humid; higher cooling demand	Urban sprawl, moderate ecological concerns	Car-dependent; limited transit coverage	High reliance on fossil fuels	Moderate water stress	Average recycling capacity
New Orleans, LA	Humid, subtropical; high cooling demand	Near wetlands; high ecological sensitivity	Car-dependent	Mostly natural gas; low renewables	High water stress	Moderate recycling systems
Inglewood, CA	Mild, Mediterranean climate	Low ecological sensitivity; urban area	Strong transit access (Metro)	High renewable energy availability	Moderate, stable supply	Advanced recycling programs
Las Vegas, NV	Desert climate; very high cooling needs	Urban sprawl; limited natural habitats	Car-dependent	Fossil-fuel heavy; some solar integration	Severe water stress	Limited recycling capabilities
Glendale, AZ	Hot, arid; very high cooling demand	Some desert ecosystems; moderate sensitivity	Car-dependent	Mostly fossil fuels; some solar	High water stress	Moderate waste processing capacity

The table provides a comparative overview of key environmental characteristics across the six candidate Super Bowl host cities—Santa Clara (CA), Atlanta (GA), New Orleans (LA), Inglewood (CA), Las Vegas (NV), and Glendale (AZ). It summarizes how each city's climate, geography, transportation infrastructure, energy mix, water availability, and waste management capacity may influence the overall environmental footprint of the event. The climate column

highlights substantial differences in expected energy demand, with desert and arid cities such as Las Vegas and Glendale requiring much more cooling energy, while coastal California cities like Santa Clara and Inglewood benefit from mild Mediterranean weather. Geographic and biodiversity conditions also vary widely: New Orleans, located near ecologically sensitive wetlands, faces higher environmental risk from large-scale events, whereas highly urbanized cities such as Santa Clara and Inglewood have lower ecological sensitivity.

Transportation infrastructure reflects another important source of variation. California cities offer well-developed public transit connections that can reduce emissions generated by spectator movement, while cities such as Atlanta, Las Vegas, and Glendale remain heavily car-dependent, increasing the likelihood of higher transportation-related carbon output. The energy mix further amplifies these differences: Santa Clara and Inglewood gain environmental advantages from access to California's renewable-rich power grid, whereas Atlanta, New Orleans, and Las Vegas rely more heavily on fossil fuels, raising their energy-related emissions. Water usage considerations also differentiate the cities; the desert regions of Nevada and Arizona experience severe water stress, while the California cities and Atlanta face only moderate pressure. The table also shows that waste management capacity varies, with Santa Clara and Inglewood demonstrating the strongest recycling and waste-handling systems, while cities like Las Vegas have comparatively limited capabilities. Overall, the table illustrates how these interconnected environmental factors create distinct sustainability profiles for each city, shaping their suitability as a host for a large-scale event such as the Super Bowl.

3.2 Variables

Key factors determining the sustainability of Super Bowl match. Many factors can contribute to the final decision of choosing an appropriate city for Super Bowl, including biological diversity, pollution, energy and resources. Therefore, to ensure that we can consider all the important factors and use these factors appropriately in determining the best city to hold Super Bowl, we use some Important criteria from the internet that people consider the most while determine sustainability.

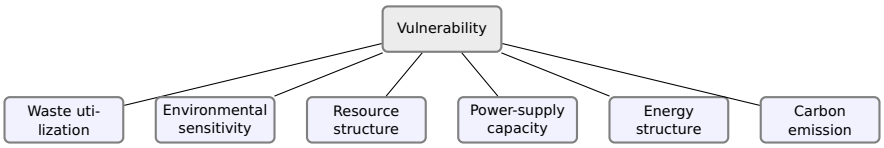


Figure 3.1: City Evaluation

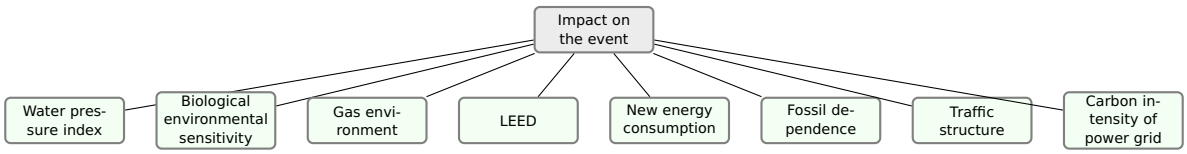


Figure 3.2: City Evaluation

3.2.1 Impact on the event: Energy structure

a) P_r Proportion of renewable energy: A quantitative, deterministic variable without units.
b) P_f Proportion of reliance on fossil energy: A quantitative, deterministic variable with no

Table 3.3: Variable Definitions

Variable Symbol	Variable Name
P_r	Proportion of renewable energy
P_f	Proportion of reliance on fossil energy
P_c	Proportion of reliance on coal
P_g	Proportion of reliance on gas
P_o	Proportion of reliance on oil
L_t	Transportation structure
S_w	Walk score
S_t	Transit score
S_b	Bike score
L_{LEED}	LEED certification level
L_g	Carbon intensity of power grid
P_w	Waste recycle rate
L_b	Biological environmental sensitivity
L_a	Air quality rank
L_w	Water stress level
L_p	Power capacity
P_e	Electricity share percentage

units. It is basically calculated by adding the percentage of coal (P_c), gas (p_g) and oil (p_o) usage.

3.2.2 Impact on the event: Carbon emission

a) L_t Transportation structure: A quantitative, deterministic variable without units. It is the geometric mean of walk score (S_w), transit score (S_t) and bike score (S_b).
b) L_{LEED} Leadership in Energy and Environmental Design certification levels: A qualitative, deterministic variable. It is a standard of measuring "green performance" of buildings, with levels including "Certified", "Silver", "Gold" and "Platinum".
c) L_g Carbon intensity of power grid: A quantitative, deterministic variable, measured in kg per kwh.

3.2.3 Urban Vulnerability: Waste treatment Capacity

a) P_w Waste recycle rate: A quantitative, deterministic variable with no units

3.2.4 Urban Vulnerability: Environmental sensitivity

a) L_b Biological environmental sensitivity: A qualitative, deterministic variable.
b) L_a Air quality rank: A quantitative, deterministic variable with no units.

3.2.5 Urban Vulnerability: Resource structure

a) L_w Water stress level: A qualitative, deterministic variable.

3.2.6 Normalization

$$P_r\text{_norm}(P_r, l, h) = \sqrt{\max\left(0, \min\left(1, \frac{P_r - l}{h - l}\right)\right)} \tag{3.1}$$

$$P_f\text{_norm}(P_f, l, h) = 1 - \sqrt{\max\left(0, \min\left(1, \frac{P_f - l}{h - l}\right)\right)} \tag{3.2}$$

$$P_{LEED}\text{_norm}(s) = \begin{cases} 1.0, & s = \text{"platinum"} \\ 0.75, & s = \text{"gold"} \\ 0.50, & s = \text{"silver"} \\ 0.25, & s = \text{"bronze"} \\ 0.0, & \text{otherwise} \end{cases} \tag{3.3}$$

$$L_t\text{_norm} = \sqrt[3]{S_w \cdot S_t \cdot S_b} \tag{3.4}$$

$$L_g\text{_norm} = 1 - \frac{\min(\max(L_g, P_5), P_{95}) - P_5}{P_{95} - P_5} \tag{3.5}$$

$$P_w\text{_norm}(P_w, l, h) = \max\left(0, \min\left(1, \frac{P_w - l}{h - l}\right)\right) \tag{3.6}$$

$$L_w\text{_norm}(s) = 1 - R(s) \tag{3.7}$$

$$R(s) = \begin{cases} 1.00, & s = \text{“extremely high”} \\ 0.75, & s = \text{“high”} \\ 0.50, & s = \text{“moderate”} \\ 0.40, & s = \text{“low-medium”} \\ 0.10, & s = \text{“low”} \\ 0.50, & \text{otherwise (default)} \end{cases} \tag{3.8}$$

$$L_b\text{_norm}(s) = M(s) \tag{3.9}$$

$$M(s) = \begin{cases} 1.00, & s = \text{“low”} \\ 0.80, & s = \text{“moderate-low”} \\ 0.60, & s = \text{“moderate”} \\ 0.40, & s = \text{“moderate-high”} \\ 0.20, & s = \text{“high”} \\ 0.60, & \text{otherwise (default)} \end{cases} \tag{3.10}$$

$$P_a\text{_norm}(P_a, l, h) = 1 - \max\left(0, \min\left(1, \frac{P_a - l}{h - l}\right)\right) \tag{3.11}$$

$$L_{p(\log)} = \ln(1 + L_p) \tag{3.12}$$

$$L_p\text{_norm} = \max\left(0, \min\left(1, \frac{L_{p(\log)} - P_5}{P_{95} - P_5}\right)\right) \tag{3.13}$$

Where: x is the original numerical value; l is the normalization lower bound; h is the normalization upper bound; Returns 0 when $h \leq l$; P is the percentile of the interval.

3.3 Modeling (Task 2)

3.3.1 Overall structure

In order to determine which city is most preferable for 2025 Super Bowl, we adopted AHP and EWM to appoint the weights of the 6 alternative choices and TOPSIS to determine the "importance" of them. These methods allow us to obtain a list of scores, which corresponds to the cities' degree of recommendation. We applied these methods for the following reasons:

3.3.2 AHP

AHP (Analytic Hierarchy Process) is a subjective multi-criteria decision-making method that combines qualitative and quantitative analysis, developed by Thomas Saaty in 1970's. It organizes complex problems into a hierarchical structure (goal, criteria, and alternatives) and uses pairwise comparisons to determine the relative importance of elements at each level.

Step1: Decision-makers assign values based on a 1–9 scale (e.g., 1 for equal importance, 9 for extreme importance) to construct a judgment matrix A,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{1n} \\ a_{21} & a_{21} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

where each element a_{ij} in A is the relative importance.

Step2: The weights of criteria are derived by calculating the maximum eigenvalue λ_{max} and the corresponding eigenvector V' of this matrix, given by normalized V'

$$AV = \lambda V$$

where V denotes the eigenvector and λ denotes the eigenvalue.
With the information given above, we could calculate the weight vector by normalize V':

$$W = (w_1, ..., w_n)$$

and

$$w_i = \frac{V_i'}{\Sigma V_i'}$$

Step3: We apply the consistency check to determine whether the weight vector is acceptable. Thus we need to calculate the followings:

$$CI = \frac{\lambda_{max} - n}{n - 1}$$
$$CR = CI/RI$$

where CI stands for consistency indicator, RI is the average random consistency indicator, which is determined by n, and CR for consistency ratio. The consistency check ensures logical reliability as consistency ratio $CR < 0.1$. For example, in evaluating the environmental impact of a Super Bowl event, AHP can hierarchically structure factors like resource consumption, waste generation, and GHG emissions, allowing experts to assign subjective weights to each criterion based on experience.

3.3.3 EWM

EWM (Entropy Weight Method) is an objective weighting method based on information entropy, which measures the uncertainty or disorder in data. It determines weights by assessing the dispersion degree of each indicator across alternatives: a smaller entropy value indicates greater variability and thus higher weight. The process involves normalizing the initial data matrix, calculating the entropy value for each indicator, and deriving weights from the entropy's divergence coefficient. In the context of Super Bowl environmental impact, EWM can objectively assign weights to indicators like energy use or transportation structure differences among cities, avoiding subjective bias from expert opinions. The steps and formulas are:

Step1: Data normalization. We first construct a matrix consisting of m alternatives and n indicators,

$$X = (x_{ij})_{m\times n} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{1n} \\ x_{21} & x_{21} & \cdots & x_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$$

and normalize it by the following transition:

$$x_{ij} \to x'_{ij} = \frac{x_{ij} - min(x_{11}, ..., x_{1n}, x_{21}, ..., x_{mn})}{max(x_{11}, ..., x_{1n}, x_{21}, ..., x_{mn}) - min(x_{11}, ..., x_{1n}, x_{21}, ..., x_{mn})}$$

and the normalized matrix

$$X' = (x'_{ij})$$

Step2: Weight and Entropy Calculation. We calculate the weight by

$$P_{ij} = \frac{P'_{ij}}{\Sigma_{i=1}^{m} P'_{ij}}$$

and entropy by

$$E_j = -\frac{1}{\ln m}\Sigma_{i=1}^{m} P_{ij} \ln P_{ij}$$

Step3: Entropy Weight calculation.

$$w_j = (1 - E_j)/\Sigma_{j=1}^{n}(1 - E_j)$$

3.3.4 TOPSIS

The TOPSIS (Technique for Order Preference by Similarity to Ideal Solution), established by C.L.Hwang and K.Yoon in 1981, ranks alternatives based on their relative proximity to ideal solutions. It identifies the positive ideal solution (PIS, with the best values across all criteria) and the negative ideal solution (NIS, with the worst values), then calculates the Euclidean distances of each alternative to these solutions. The alternative with the smallest distance to the PIS and largest distance to the NIS is preferred. TOPSIS is effective for small sample sizes and requires no assumptions about data distribution. For Super Bowl analysis, it can rank cities by environmental performance. (e.g., comparing their resource use and emissions to an "ideal" green city) The steps include:

Step1: Data normalization. Since our goal for different indicators might be different (Maximization, Minimization, Approaching to some values or intervals), we have to transform these goals into maximization. We denote the matrix consisting of m alternatives and n indicators as $X = (x_{ij})_{m\times n}$
For minimization, we could accomplish this by the transition:

$$x_i \to x_i' = max - x_i$$

For an approach to a certain value k, we could apply:

$$x_i \to x_i' = 1 - \frac{x_i - k}{max(|x_i - k|)}$$

For an approach to a certain interval $[a, b]$, we use:

$$x_i \to x_i' = \begin{cases} 1 - \frac{a - x_i}{M}, & x_i < a \\ 1, & a < x_i < b \\ 1 - \frac{x_i - b}{M}, & x_i > b \end{cases}$$

where

$$M = \max\{a - \min\{x_i\}, \max\{x_i\} - b\}$$

Step2: Data standardization. After the transition $X \to X'$, the data has been normalized, but to eliminate the influence of different data indicator dimensions, it is also necessary for us to standardize the already normalized matrices by

$$x_{ij}' \to r_{ij} = \frac{x_{ij}'}{\sqrt{\sum_{i=1}^{n} x_{ij}'^2}}$$

$$X' \to R$$

Step3: The rating matrix Z, which has undergone both normalization and standardized processing, contains "maximization" data. Then we can extract the ideal optimal solution and the worst solution from it. (PIS and NIS, as shown above) Therefore, we take out each indicator, that is, the largest number in each column, to form the ideal optimal solution vector, that is

$$z^+ = [z_1^+, z_2^+, ..., z_m^+] = [\max\{z_{11}, z_{21}, ..., z_{n1}\}, \max\{z_{12}, z_{22}, ..., z_{n2}\}, ..., \max\{z_{1m}, z_{2m}, ..., z_{nm}\}]$$

Similarly, take the smallest number in each column to calculate the ideal worst solution vector

$$z^- = [z_1^-, z_2^-, ..., z_m^-] = [\min\{z_{11}, z_{21}, ..., z_{n1}\}, \min\{z_{12}, z_{22}, ..., z_{n2}\}, ..., \min\{z_{1m}, z_{2m}, ..., z_{nm}\}]$$

Step5: Calculate the Euclidean distances between every possible solution to PIS and NIS:

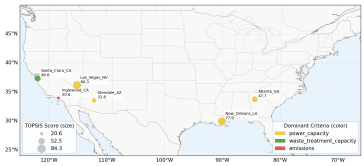
$$d_i^+ = \sqrt{\sum_{j=1}^{m} (z_j^+ - z_{ij})^2}$$

$$d_i^- = \sqrt{\sum_{j=1}^{m} (z_j^- - z_{ij})^2}$$

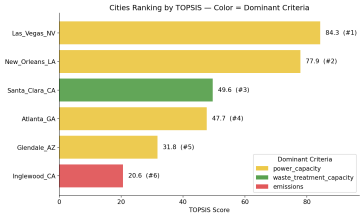
Step6: We use the evaluation function

$$S_i = \frac{d_i^-}{d_i^+ + d_i^-}$$

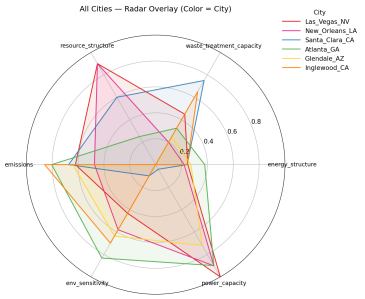
to evaluate solutions. It is clear that $0 \leq S \leq 1$ and as d_i^+ decreases (The distance between this solution and the optimal solution is smaller), S_i gets closer to 1, and vise versa.



(a) Map & Score



(b) Ranking



(c) Cities Comparison

3.4 Model Application (Task 3)

3.4.1 Application 1: City Recommendation

Based on the integrated weighting framework and the subsequent TOPSIS evaluation, the model identifies Las Vegas, Nevada as the most environmentally favorable host city under the updated indicator set. Las Vegas achieves the highest overall performance score driven by its strong energy structure, comparatively low ecological sensitivity, and particularly robust power capacity. Its position at the top of the ranking indicates that the expected environmental pressure associated with large-scale events would be significantly reduced compared with the alternative cities evaluated.

New Orleans, Louisiana emerges as the second-ranked city. New Orleans demonstrates relatively balanced performance across most environmental categories, benefiting especially from favorable energy structure conditions and high ecological sensitivity. Its high scores in these areas compensate for moderate emissions and waste-treatment capacities. However, while New Orleans performs strongly in several aspects, its overall environmental burden indicators do not surpass those of Las Vegas, preventing it from taking the leading position.

Ranked third, Atlanta, Georgia shows competitive results in multiple criteria, including energy structure and power capacity. Nevertheless, Atlanta faces challenges in emissions and ecological sensitivity, where its performance lags behind the two leading cities. These limitations weaken its overall sustainability profile and reduce its competitiveness relative to Las Vegas and New Orleans.

The remaining candidate cities exhibit lower composite scores, reflecting more pronounced environmental constraints. Santa Clara, California, despite previously strong performance in earlier evaluations, shows notable weaknesses in power capacity and emissions under the updated indicator set, significantly diminishing its relative ranking. Glendale, Arizona displays moderate performance across resource and environmental sensitivity categories but lacks strength in waste-treatment capacity. Inglewood, California, positioned last, demonstrates substantial weaknesses in power capacity, emissions, and overall resource structure, resulting in the lowest TOPSIS closeness value among all candidates.

In summary, the updated model robustly recommends Las Vegas as the most environmentally suitable option under the revised evaluation criteria. Its high performance across key weighted indicators positions it closest to the ideal environmental profile and suggests that it would impose the lowest environmental impact among the evaluated candidate cities.

City Name	Energy Structure	Waste Treatment	Resource Structure	Emissions	Environmental Sensitivity	Power Capacity	TOPSIS Score	Rank
Las_Vegas_NV	0.2311	0.45	0.900	0.6335	0.4337	1.0000	83.8997	1
New_Orleans_LA	0.2125	0.20	0.900	0.5021	0.5805	0.9022	76.8555	2
Atlanta_GA	0.3663	0.325	0.250	0.8053	0.8326	0.8996	49.3858	3
Santa_Clara_CA	0.2318	0.75	0.600	0.7118	0.1000	0.0410	48.1207	4
Glendale_AZ	0.2537	0.25	0.000	0.6629	0.6366	0.7209	33.7207	5
Inglewood_CA	0.2318	0.65	0.000	0.8776	0.7000	0.0000	21.1769	6

The following figures visually demonstrate the results. In Figure 3.3, the six cities included in the ranking are shown on the map of the continental United States. The size of the city symbols indicates the relative magnitudes of the TOPSIS scores. The colors of the symbols represent the dominant criteria in each city. In Figure 3.4, the bar graph displays the ranking of the cities based on the TOPSIS scores more directly, indicating that Las Vegas in Nevada has the highest recommendation priority, while the colors still correspond to the dominant criteria in each city. In Figure 3.5, the radar map illustrates the differences between the six cities, allowing us to directly identify which city performs best on each individual factor.

3.4.2 Application 2: Considering Three More Cities

Based on the integrated weighting scheme and the updated TOPSIS evaluation, the model identifies Jacksonville, Florida as the most environmentally suitable candidate among the assessed cities. Jacksonville achieves the highest composite performance score, supported by a balanced energy structure, moderate emissions levels, and strong environmental sensitivity. Notably, its full score in power capacity significantly strengthens its overall sustainability profile, suggesting that its environmental burden during large-scale events would be relatively low compared with the competing locations.

Cincinnati, Ohio emerges as the second-ranked city. Cincinnati performs strongly across several environmental indicators, particularly in resource structure, where it outperforms all other candidates. Although its emissions level is higher than that of Jacksonville, its solid performance in environmental sensitivity and substantial power capacity bolster its overall ranking. Nevertheless, the marginal differences in key indicators prevent Cincinnati from surpassing Jacksonville's leading environmental position.

Ranked third, Houston, Texas shows considerable limitations across multiple dimensions. While Houston demonstrates moderate resource structure and acceptable emissions performance, its significantly reduced power capacity and relatively low waste-treatment performance weaken its overall environmental resilience. These constraints result in the lowest TOPSIS score among the evaluated cities, indicating a substantial deviation from the ideal environmental profile.

In summary, the updated model robustly recommends Jacksonville as the most environmentally favorable option. Its balanced performance across weighted sustainability indicators positions it closest to the ideal solution, indicating that it would impose the lowest expected environmental impact under the criteria established in this study.

City Name	Energy Structure	Waste Treatment	Resource Structure	Emissions	Environmental Sensitivity	Power Capacity	TOPSIS Score	Rank
Jacksonville_FL	0.2902	0.2740	0.250	0.4079	0.7000	1.0000	75.5275	1
Cincinnati_OH	0.2571	0.1920	0.900	0.6450	0.5330	0.7389	75.3639	2
Houston_TX	0.1721	0.1410	0.500	0.4940	0.5000	0.0000	10.8764	3

3.5 Model Expansion: World Cup (Task 4)

3.5.1 Model Revision

Considering the differences between the Superbowl and individual-sport championship tournaments, our team identifies two main differences among various single-sport championship tournaments. The first is international viewership. The Superbowl is held in the United States, but most of its audience comes from the U.S., making its international reach and viewership lower than other tournaments. For example, the World Cup's audience spans the globe, across every continent and country, so its international influence far surpasses that of the Superbowl. Because of this, people travel from around the world to attend generate significantly more CO_2 due to flying, which leads to higher emissions compared to other transportation methods. In contrast, the Superbowl's audience mainly resides in the U.S., resulting in less CO_2 emissions from attendees. Therefore, when selecting a city for a single-sport championship, we should include accessibility as a criterion. The second difference concerns environmental impact. Some tournaments affect water resources more, while others contribute more to CO_2 emissions. Considering these differences, we adjust the weights of various criteria to better match the specific tournament.

Basically, we revise the model by adding a variable, accessibility, which can be represented by A. A_w represents the weight of potential audience from different continents, which can be expressed as:

$$A_w(s) = \begin{cases} 0.40, & s = \text{“Asia”} \\ 0.30, & s = \text{“Europe”} \\ 0.125, & s = \text{“Americas”} \\ 0.10, & s = \text{“Africa”} \\ 0.025, & s = \text{“Oceania”} \end{cases} \tag{3.14}$$

Besides, we consider a specific city's air connectivity efficiency, calculated as a function of flight time and aviation connectivity index(CI) (Table 3.3):

Table 3.4: Flight duration (in hours) from major cities to different continents

City	CI	To Asia	To Europe	To America	To Africa	To Oceania
Beijing	1.0	4 hours	10 hours	14 hours	12 hours	11 hours
London	1.0	11 hours	2 hours	9 hours	8 hours	22 hours
Paris	1.0	11 hours	1.5 hours	9 hours	6 hours	22 hours
Toronto	0.8	13 hours	7 hours	3 hours	12 hours	18 hours
Sydney	0.8	10 hours	22 hours	16 hours	14 hours	3 hours
Buenos Aires	0.7	24 hours	12 hours	10 hours	9 hours	12 hours

The total score of accessibility is then calculated, using the following formula, where t_{norm} is ratio between the time of flight and 10 :

$$A = \sum_i \left[\mathrm{A}_{\mathrm{W}i} \times \left(\frac{1}{t_{\mathrm{norm},i}} \times \mathrm{CI} \right) \right] \tag{3.15}$$

The results yield a clear ranking: Paris (1.94) and London (1.81) have the optimal accessibility score, since they are the central location within Europe, which is the largest audience market, and their status as global aviation hubs. Beijing (1.50) ranks third due to the high weight Asian market. Toronto (1.16), Buenos Aires (0.70), and Sydney (0.61) follow, as they are relatively isolated from each other places. This analysis provides a critical, data-driven dimension for evaluating a city's capability to minimize global travel barriers for a worldwide sporting event.

The steps of model establishment are basically the same as those we did in task 2. However, the weight of some factors that we gained from AHP varied a little bit, and the next part is the model application.

3.5.2 Revised Model Application

Based on the integrated weighting framework and the subsequent TOPSIS evaluation for the "WorldCup" profile, the model identifies Beijing, China as the most environmentally favorable host city. Beijing achieves the highest overall TOPSIS score of 76.26, driven by its exceptional performance in accessibility (0.850) and perfect power capacity (1.000), complemented by strong emissions control (0.630). Its position at the top of the ranking indicates that the expected environmental pressure associated with hosting a large-scale global event would be significantly reduced compared to the alternative cities evaluated.

London, United Kingdom emerges as the second-ranked city with a score of 67.67. London demonstrates outstanding performance in accessibility (0.973) and emissions (0.826), while also showing considerable strength in environmental sensitivity (0.658). These advantages position it as a highly competitive candidate, though its relatively lower power capacity (0.260) prevents it from overtaking Beijing for the top position.

Ranked third with a score of 66.14, Paris, France achieves perfect accessibility (1.000) and excellent emissions performance (0.858). However, Paris shows relative weaknesses in power capacity (0.205) and waste treatment capacity (0.200) compared to the top two cities, which slightly diminishes its overall sustainability profile.

The remaining candidate cities exhibit more pronounced environmental constraints. Toronto, Canada (score: 37.24) shows moderate performance across multiple criteria but is limited by its accessibility (0.424) and minimal power capacity (0.041). Sydney, Australia (score: 14.14) and Buenos Aires, Argentina (score: 11.18) face significant challenges, particularly in accessibility (0.093 and 0.000 respectively) and power capacity, which substantially diminish their competitiveness despite moderate performances in some environmental categories.

In summary, the model robustly recommends Beijing as the most suitable host under the WorldCup evaluation criteria. Its superior performance across the most heavily weighted indicators—particularly accessibility (weight: 0.2599), power capacity (weight: 0.1692), and emissions (weight: 0.2258)—positions it closest to the ideal environmental profile and suggests it would impose the lowest environmental impact among the six global cities evaluated for this major sporting event.

Table 3.5: City Environmental Assessment Results

City	Accessibility	Energy Structure	Emissions	Resource Structure	Waste Treatment	Env Sensitivity	Power Capacity	TOPSIS Score	Rank
Beijing	0.850	0.250	0.630	0.250	0.250	0.200	1.000	76.26	1
London	0.973	0.378	0.826	0.500	0.330	0.658	0.260	67.67	2
Paris	1.000	0.421	0.858	0.500	0.200	0.603	0.205	66.14	3
Toronto	0.424	0.396	0.744	0.500	0.520	0.700	0.041	37.24	4
Sydney	0.093	0.264	0.652	0.250	0.400	0.447	0.096	14.14	5
Buenos Aires	0.000	0.284	0.597	0.500	0.150	0.592	0.000	11.18	6

3.5.3 Strategies Analysis

This section provides some strategies for cities to enhance their sustainability profile, so they can be more competitive in holding those international sports events. Based on the model outcomes, we propose and stimulate three targeted intervention strategies. Our analysis focuses on a mid-tier and a lower-tier city, which are Toronto in Rank 4 and Buenos Aires in Rank 6. The cases can serve as an example to demonstrate the varying efficacy of strategies across different developmental contexts. In our model, infrastructure investment mainly affects waste treatment

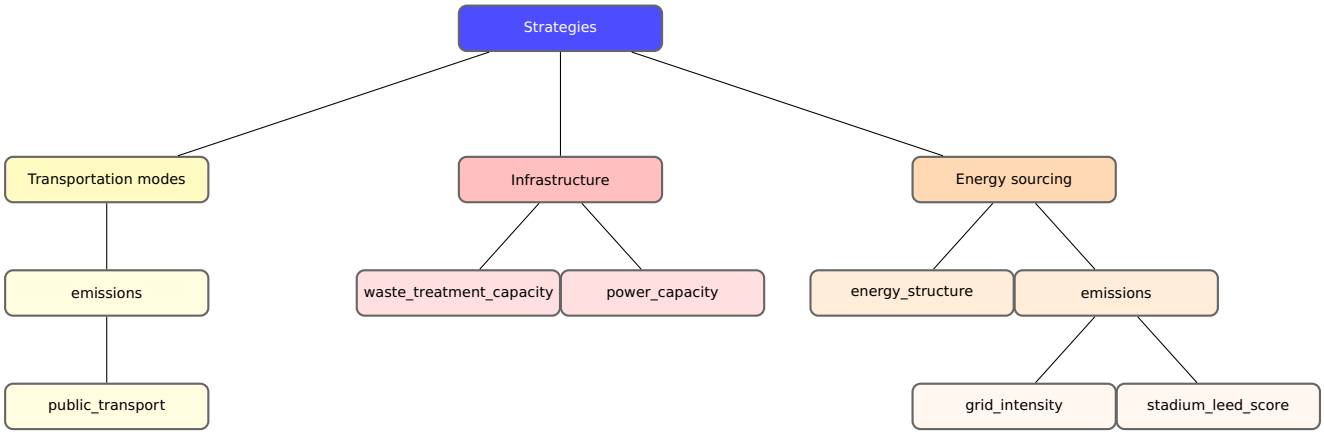


Figure 3.4: Strategic Analysis Framework

capacity and power capacity. In the short term, cities do not need to build entirely new systems; they can expand and modernize existing plants, improve recycling and food-waste separation at stadiums, and reinforce substations and local storage around venues. These measures quickly reduce uncontrolled landfill, low-standard incineration and blackout risk during the event, and they translate directly into higher waste_treatment_capacity and power_capacity scores. Because they are upgrades and targeted reinforcements rather than Greenfield megaprojects, the capital cost and construction time remain manageable, making this type of infrastructure investment both environmentally effective during the tournament and economically and logistically feasible within a single bidding cycle.

Energy-sourcing policies in our framework change the energy_structure and emissions-related indicators, especially the renewable share, the dependence on fossil fuels, the carbon intensity of the grid, and the "greenness" of stadiums. In the short run, host candidates do not need to decarbonize their entire economy; they can focus on the electricity and fuels used directly by venues and broadcasting facilities. Signing power purchase agreements for renewable electricity, buying credible renewable energy certificates, and retrofitting stadiums with efficient lighting, HVAC and, where possible, rooftop solar all raise the energy-related scores without requiring a complete national energy transition. Because PPAs and certificates are contractual tools and stadium retrofits can be aligned with scheduled maintenance, these energy measures are realistic pre-event actions that substantially lower event-related emissions

while staying within political and financial constraints.

Transport policy in the model primarily improves the public-transport and accessibility components of the emissions dimension. Cities can cut car dependence around venues by increasing metro and bus frequency on match days, adding shuttle services from park-and-ride hubs and main railway stations, bundling match tickets with free or discounted public transport, and using temporary traffic management such as car-free streets or reduced parking near stadiums. These are mostly operational measures, relying on existing networks and temporary infrastructure, so they can be deployed quickly and at relatively low cost. In the model, such changes increase the public_transport indicator and reduce transport-related emissions and local air pollution during the tournament, while being logistically feasible for transport authorities that already manage peak-hour traffic.

To illustrate how these levers affect long-term hosting desirability, we compare Toronto, which currently ranks fourth in our six-city sample, with Buenos Aires, which ranks sixth. Toronto already has relatively strong performance across criteria; if it systematically directs more renewable electricity to venues, pushes stadium efficiency upgrades, and institutionalizes event-linked public transport measures, our simulations show its composite score rising and the gap to London and Paris narrowing, making it a consistently "safe and sustainable" choice in future bidding cycles. Buenos Aires starts from a lower baseline, with weaker waste management, power capacity, and energy structure, but this also means it has more headroom for improvement. If Buenos Aires concentrates investment on modern waste and recycling facilities in stadium districts, upgrades local grid reliability, adopts a clear renewable-electricity strategy for events, and strengthens mass transit operations around matches, its scores in our model increase enough to move it out of last place and closer to the middle of the group. This suggests that targeted, feasible policies in infrastructure, energy, and transport can both reduce short-term environmental impacts and significantly improve a city's long-term attractiveness as a sustainable host, especially for currently lower-ranked candidates like Buenos Aires.

3.5.4 Main differences in sustainability impacts

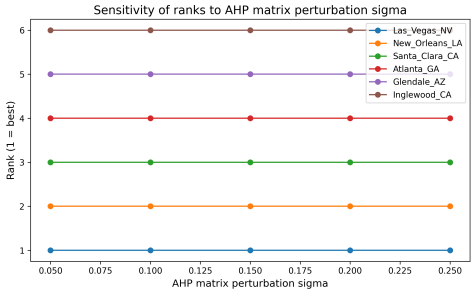
Although both single-sport mega-events like the Super Bowl and multi-sport spectacles like the Olympics face sustainability challenges, their core environmental impacts differ fundamentally in scope and nature. A single-game event generates an intense, short-term spike in operational pressures—its primary footprint stems from Scope 3 emissions, particularly spectator travel, and resource consumption concentrated within the event period. In contrast, a multi-sport event raises a long-term, systemic environmental question, where the embodied carbon from constructing new venues and infrastructure often outweighs operational emissions. The former tests a city's ability to manage a temporary surge efficiently, while the latter tests its vision and capacity for sustainable urban transformation—leaving behind legacy infrastructure that may become either a lasting asset or a long-term environmental liability. These differences in sustainability impact are further illustrated in the figure below.

Table 3.6: Comparing Sustainability Impacts: Single-Sport Events vs. Multi-Sport Events

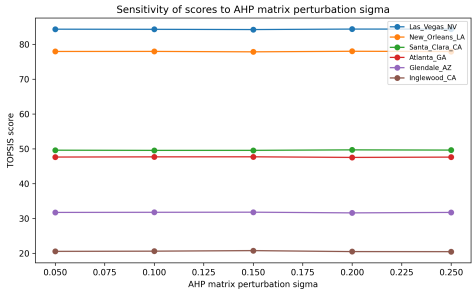
Impact Dimension	**Single-Sport Events (e.g., Super Bowl)**	**Multi-Sport Events (e.g., Olympics)**
Core Impact	Short-term, concentrated operational impact	Long-term, profound systemic impact
Carbon Emission Sources	Primarily Scope 3 emissions (mainly spectator travel)	Embodied + operational carbon (venue construction + systemic operations)
Resource Management	Short-term resource consumption peaks (energy, water, single-use items)	Long-term resource commitment
Environmental Risks	Acute, localized (traffic congestion, temporary pollution)	Chronic, systemic (venue underutilization, ecological disruption)

3.6 Sensitivity Analysis

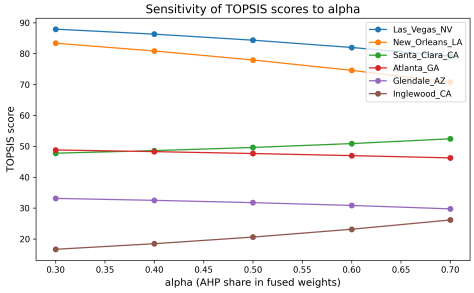
We applied multi-seed sensitivity analysis to our model. The analysis was carried out with a sample size of 1000 and averaged across 5 different random seeds (11, 23, 37, 61, 89). In the analysis, the model hyperparameters were perturbed within specific ranges:



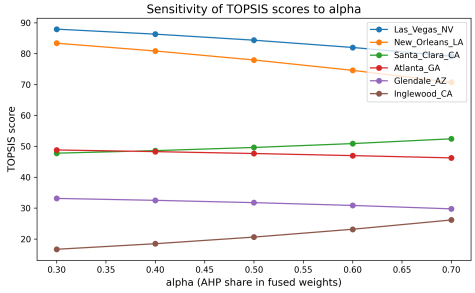
(a) Change in Rank with the AHP matrix perturbation sigma



(b) Change in TOPSIS scores with the AHP matrix perturbation sigma

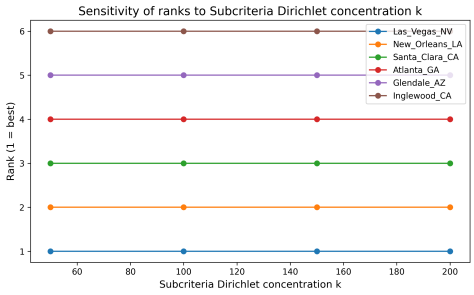


(a) Change in Rank with the AHP share in fused weights

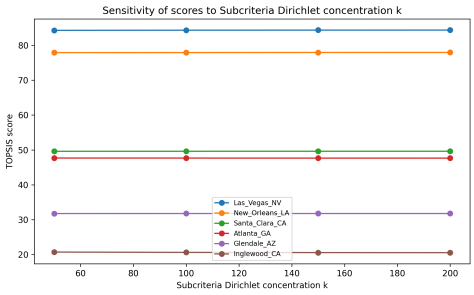


(b) Change in TOPSIS scores with the AHP share in fused weights

In these pictures, the AHP matrix perturbation sigma is the hyperparameter which simulates the uncertainty in judgment and was perturbed from 0.050 to 0.250 with a default value of



(a) Change in Rank with the subcriteria dirichlet concentration k



(b) Change in TOPSIS scores with the subcriteria dirichlet concentration k

0.150. The AHP share in fused weights is the hyperparameter which represents the ratio of AHP in the total weights (AHP+EWM) and was perturbed from 0.20 to 0.50 with a default value of 0.30. The subcriteria dirichlet concentration k is the hyperparameter which represents the perturbation of sub-weights and was perturbed from 50.0 to 200.0 with a default value of 100.0.

We also applied the same method to test our model with the extra 3 cities and the expanded model for World Cup.

Based on the sensitivity analysis results, the model demonstrates strong robustness. The multi-seed analysis reveals that perturbations to the three key hyperparameters—AHP matrix perturbation sigma, AHP share in fused weights, and subcriteria dirichlet concentration k—within their specified ranges have only a minimal impact on the model's outputs. The changes in both the final rankings and the TOPSIS scores are relatively stable and small across the tested values for each hyperparameter. This indicates that the model's decisions are not highly sensitive to variations in the uncertainty of expert judgments (simulated by the AHP matrix perturbation), the balance between AHP and EWM weights, or the perturbation of sub-criteria weights. Consequently, the model can be considered reliable and consistent under different parameter configurations.

3.7 Strategies and their improvements on cities

In our model, infrastructure investment mainly affects waste treatment capacity and power capacity. In the short term, cities do not need to build entirely new systems; they can expand and modernize existing plants, improve recycling and food-waste separation at stadiums, and reinforce substations and local storage around venues. These measures quickly reduce uncontrolled landfill, low-standard incineration and blackout risk during the event, and they translate directly into higher waste treatment capacity and power capacity scores. Because they are upgrades and targeted reinforcements rather than Greenfield mega-projects, the capital cost and construction time remain manageable, making this type of infrastructure investment both environmentally effective during the tournament and economically and logistically feasible within a single bidding cycle.

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3.8 Advantages and Limitations

3.8.1 Advantages

1. Integration of Subjective and Objective Weighting The model combines the subjective expert judgment of AHP with the objective data-driven approach of EWM, balancing expert insight with statistical evidence. This integrated method reduces potential biases inherent in either single method, resulting in a more scientific and convincing weighting system.

2. Comprehensive and Structured Evaluation System A multi-dimensional indicator system is constructed, assessing both the direct environmental pressure from event operations (e.g., carbon emissions) and the intrinsic environmental vulnerability of the city (e.g., water stress). This dual perspective ensures a systematic and in-depth evaluation.

3.8.2 Limitations

1. Dependence on Data Quality and Simplifying Assumptions

The model's effectiveness relies heavily on the accuracy and completeness of input data. While necessary due to data availability, certain assumptions—such as using population-weighted state-level estimates, applying past event impact data, and using PM2.5 as a proxy for overall air quality—may introduce bias. Future efforts to collect more granular local data and incorporate dynamic environmental impact factors could enhance model precision.

2. Limitations of Static Weight Allocation The model employs fixed weights, which may not adequately capture the nonlinear characteristics and threshold effects of certain environmental indicators. For instance, when a city's water stress exceeds a critical level, its importance should increase significantly—a dynamic response that static weights cannot reflect. Exploring adaptive weighting mechanisms or incorporating constraint-based evaluations could be valuable future enhancements.

AI report

In the process of writing this paper, generative AI tools were not utilized for core aspects such as model construction, data analysis, or content drafting. All research outcomes presented are entirely the result of the authors' independent work.

4 Letter to NFL leadership(Task 5)

Dear NFL Leadership,

We have conducted a systematic evaluation of potential host cities for Super Bowl LXIII in 2029 using our independently developed assessment model, with a focus on environmental sustainability. This evaluation strictly adheres to the NFL's sustainability indicator system, emphasizing key environmental factors such as energy consumption, water usage, waste management, and greenhouse gas emissions. Our assessment framework is structured around two dimensions: first, analyzing the impact of a city's infrastructure green level, resource efficiency, and environmental management capabilities on the event's environmental performance; and second, evaluating the city's ecological sensitivity and capacity to respond to environmental changes, thereby comprehensively assessing its environmental resilience.

Based on our model evaluation results, Las Vegas has demonstrated outstanding performance among the candidate cities, emerging as the preferred choice to host Super Bowl LXIII in 2029. Below, we elaborate on its advantages in three key areas:

First, Las Vegas possesses significant advantages in power supply capacity. Its robust grid infrastructure and stable power supply capability ensure that the high electricity demands during the Super Bowl can be fully met without placing excessive strain on the local grid system. This characteristic aligns perfectly with the NFL's sustainability goals, which prioritize energy reliability and efficiency.

Second, Las Vegas excels in water stress indicators. Data shows that the city's water stress index remains below 10 percent, indicating a secure water resource level. This means that the water consumption associated with the event will not significantly impact the local water resource system. This strength is highly consistent with the NFL's sustainability objectives for water resource conservation and efficient utilization.

Lastly, Las Vegas also demonstrates advantages in carbon emission control. As a LEED Gold-certified venue, Allegiant Stadium meets high standards in energy efficiency, material usage, and indoor environmental quality. Additionally, the local grid's low carbon intensity ensures relatively low carbon dioxide emissions from electricity generation. These features collectively guarantee that hosting the Super Bowl in Las Vegas will result in minimal greenhouse gas emissions, aligning with the NFL's commitment to addressing climate change.

In summary, Las Vegas performs exceptionally well across the three critical dimensions of energy security, water resource safety, and carbon emission control, making it an ideal choice to host Super Bowl LXIII in 2029. We are confident that selecting Las Vegas will contribute to achieving the NFL's strategic environmental sustainability goals while setting a new benchmark for eco-friendly standards in this premier global sporting event.

Sincerely, On behalf of the 2029 Super Bowl Sustainability Analysis Team

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